ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO4 – Investigative Assignment

Ultra-High-Speed Photonic Transport Link: Technology Evaluation, Impairment Mitigation, and Integrated System Design

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Register Number	192312050
Course Outcome	CO4
Nature of Assessment	Investigative Assignment
Maximum Marks	100
Bloom's Levels	K4 – Analyze | K5 – Evaluate | K6 – Create/Design
Knowledge Profile	WK1, WK3, WK5

Engineering Investigation Problem

A next-generation optical network operator plans to deploy a 32-channel DWDM transport link, each channel carrying 200 Gb/s, compliant with ITU-T G.698 grid spacing and IEEE 802.3ct/ck / OIF-CEI standards. This report investigates candidate modulation formats, modulator mechanisms/architectures, optical amplification strategies, and coherent DSP impairment-compensation schemes, quantifies their trade-offs, and proposes a final integrated end-to-end system design, following the workflow: REQUIREMENT → LITERATURE → ALTERNATIVES → ANALYSIS → SELECTION → SYSTEM DESIGN → VERIFICATION → FINAL DECISION.

Given Design Constraints

Parameter	Value
WDM configuration	32-channel DWDM, ITU-T G.698 grid
Target channel rate	200 Gb/s per channel
Candidate modulation formats	OOK(NRZ/RZ), PAM4, DPSK, DQPSK, QAM, Coherent DP-16/64QAM
Candidate modulator mechanisms	EO, AO, thermo-optic, magneto-optic, Kerr; MZI, MRR, EO polymer, Si-photonic
Candidate amplifiers	EDFA, SOA, Distributed/Lumped Raman, Hybrid (EDFA+Raman)
Amplifier span condition	Input –15 dBm/ch (32 ch), target output +2 dBm/ch
Target pre-FEC BER	< 3.8 × 10■³
Target OSNR	> 18 dB
Target spectral efficiency	≥ 4 b/s/Hz
Target modulator bandwidth	> 35 GHz, compact PIC footprint

STAGE 1 – Investigation Requirement Statement (15 Marks)

1.1 Literature and Standards Context

ITU-T G.698 defines the DWDM grid, launch-power and OSNR application-code framework for multichannel amplified systems; IEEE 802.3ct/ck and OIF-CEI define the electrical/optical interface, baud-rate and FEC conventions for 100G-per-lane and beyond coherent/PAM links used in current 400G–800G ZR/ZR+ and data-centre interconnect deployments. Across recent literature, coherent DP-QAM formats with digital signal processing (DSP) are the dominant choice for 200 Gb/s-class per-channel long-haul/metro DWDM, while PAM4 with direct detection remains the standard for shorter-reach, lower-cost intra-/inter-data-centre links — this divide motivates the Stage 2 comparison below.

1.2 Aggregate Capacity

Aggregate capacity = 32 channels × 200 Gb/s/channel = ***6.4 Tb/s***

1.3 Key Physical-Layer Metrics to Investigate

- **Spectral efficiency (b/s/Hz):** bits transmitted per unit optical bandwidth — governs how many 200 Gb/s channels fit in the available C-band grid.
- **OSNR requirement:** minimum optical SNR (in a reference bandwidth) needed to achieve target BER; rises sharply with modulation order.
- **BER / Q-factor:** end-to-end transmission-quality metrics; Q-factor (dB) is a convenient linear proxy for pre-FEC BER.
- **Optical launch power per channel:** must be high enough for good OSNR but below the nonlinear (SPM/XPM/FWM) threshold.
- **Modulation bandwidth:** modulator and receiver electrical bandwidth must support the required baud rate.

- **Insertion loss:** of modulators, AWG/MMI multiplexers and isolators, which sets the power budget.
- **Thermal / footprint constraints:** especially relevant for compact photonic-integrated-circuit (PIC) transceivers.

1.4 Impairments Mapped to DSP Functions

Impairment	Physical Origin	Primary Mitigation
Chromatic Dispersion (CD)	Wavelength-dependent group velocity in fiber	Static FIR / frequency-domain equalizer (DSP)
Polarization Mode Dispersion (PMD)	Birefringence, random polarization coupling	Adaptive MIMO equalizer (e.g., CMA)
SPM / XPM / FWM (Kerr nonlinearities)	Intensity-dependent refractive index	Digital backpropagation (DBP); power/dispersion-map c
WDM crosstalk	Imperfect AWG/MMI channel isolation	Guard-band design, filter roll-off control
Amplifier ASE noise	Spontaneous emission in optical amplifiers	Optimum amplifier spacing/type, low-NF design

1.5 Practical Trade-offs

Silicon Photonics offers the best CMOS-compatible integration density and lowest cost at scale, but with higher Vπ and thermal sensitivity than Lithium Niobate (LiNbO■), which offers superior linearity and low chirp at the cost of larger device size and higher assembly cost; EO-polymer modulators promise very low Vπ and high bandwidth but currently lag in long-term reliability. Power consumption and reliability generally favour higher levels of photonic integration, provided thermal management is addressed.

Stage 1 Deliverable – Investigation Requirement Statement

The proposed DWDM transport link must deliver 6.4 Tb/s aggregate capacity across 32 channels at 200 Gb/s/channel within the ITU-T G.698 grid, achieving ≥4 b/s/Hz spectral efficiency, >18 dB OSNR and pre-FEC BER <3.8×10■³, using a modulator with >35 GHz bandwidth and compact PIC footprint, an amplification chain that limits cumulative ASE-driven OSNR penalty, and a coherent DSP chain that fully compensates CD, PMD and nonlinear impairments — all while remaining compliant with IEEE 802.3ct/ck / OIF-CEI interface standards.

STAGE 2 – Modulation / Modulator Configuration (20 Marks)

2.1 Modulation Format Comparison

Format	Bits/Symbol	Spectral Efficiency*	SNR for 10■³ BER	Impairment Tolerance
NRZ-OOK	1	~1 b/s/Hz	Low requirement, but poor noise margin	Poor (no coherent compensation)
PAM4	2	~2 b/s/Hz	~4–5 dB higher than OOK for same BER	Moderate; needs strong FEC/equaliza
DPSK	1	~1 b/s/Hz	~3 dB better than OOK (differential detection gain)	Moderate
DQPSK	2	~2 b/s/Hz	Better than DPSK per bit, still direct-detection	Moderate
16-QAM (single pol, coherent)	4	~4 b/s/Hz	Higher OSNR requirement than QPSK	Good, with coherent DSP
Coherent DP-16QAM	8 (4×2 pol)	~8 b/s/Hz (ideal)	~16–17 dB OSNR (0.1nm ref) for ~1E■ post-FEC BER	Excellent—full linear+MIMO DSP

*Ideal, dual-polarization-aware values before implementation/roll-off/overhead penalties.

2.2 Baud-Rate and Channel-Spacing Calculation for 200 Gb/s

DP-16QAM: 16-QAM carries 4 bits/symbol; dual-polarization transmission doubles this to 8 bits/symbol per unit time.

Baud rate = Bit rate / bits-per-symbol = 200 Gb/s / 8 = ***25 Gbaud*** *(ideal, no overhead)*

With ~20% FEC/framing overhead: 200×1.2 / 8 ≈ ***30 Gbaud***

At ~30 Gbaud with Nyquist pulse shaping, the occupied optical bandwidth is approximately 1.1–1.5 × baud rate ≈ 33–45 GHz, which comfortably fits within a **50 GHz ITU-T G.698 grid slot** (32 × 50 GHz = 1.6 THz, well inside the ~4.4 THz usable C-band).

PAM4 (direct detection, single polarization): 2 bits/symbol.

Baud rate = 200 Gb/s / 2 = ***100 Gbaud*** *(ideal); with 20% overhead ≈* ***120 Gbaud***

A 100–120 Gbaud PAM4 signal needs roughly 60–75 GHz of electrical/optical bandwidth — this exceeds a 50 GHz DWDM grid slot and pushes modulator/driver/ADC bandwidth requirements (and dispersion sensitivity) to levels that are impractical for 32-channel long-reach DWDM, even though PAM4 is simpler and lower-cost for short-reach direct-detection links.

2.3 Modulation-Mechanism Comparison

Mechanism	Speed Limit	Vπ	Response Time	Chirp
Electro-optic (EO, Pockels effect)	Very high (>40 GHz, travelling-wave)	Low–moderate (LiNbO■, P■ZT, Si-hybrid)	Picosecond-scale	Low (can be designed near-zero ch
Acousto-optic (AO)	Low (MHz range, limited by acoustic transit time)	N/A (RF drive power based)	Microsecond-scale	Not applicable in the same sense
Thermo-optic	Very low (kHz–MHz, thermal diffusion limited)	N/A (power, not voltage, driven)	Microsecond–millisecond	Negligible (slow, phase-only)
Magneto-optic	Low–moderate	N/A	Moderate	Low
Kerr-effect (all-optical)	Very high (fs–ps, ultrafast)	N/A (intensity-driven)	Femtosecond–picosecond	Can be significant (self-phase mod

The **electro-optic effect** is the only mechanism in this list combining the very high speed (>35–40 GHz) and low, controllable chirp needed for coherent DP-16QAM at 25–30 Gbaud; thermo-optic and acousto-optic mechanisms are two to six orders of magnitude too slow, and pure Kerr-effect all-optical modulation, while fast, is harder to control for linear, low-chirp high-order QAM encoding.

2.4 MZI vs Micro-Ring Resonator (MRR) Comparison

Parameter	Mach-Zehnder Interferometer (MZI)	Micro-Ring Resonator (MRR)
Bandwidth	Very high, broadband (travelling-wave design)	High but resonance-limited; narrower native modulation bandw

Parameter	Mach-Zehnder Interferometer (MZI)	Micro-Ring Resonator (MRR)
Footprint	Larger (mm-scale in Si; cm-scale in LiNbO■)	Very compact (µm-scale ring radius)
Thermal sensitivity	Low–moderate	High (resonance wavelength shifts strongly with temperature, r
Insertion loss	Low–moderate, well-controlled	Can be low at resonance but sensitive to fabrication/thermal dr
Linearity / chirp control	Excellent – well-established push-pull biasing for near-zero chirp	More complex to achieve equivalent linearity
Best fit	High-order QAM, long-haul coherent transmission	Dense WDM transmitter arrays where ultra-compact footprint d

Design Decision 1 – Modulation / Modulator Configuration

I recommend **Coherent DP-16QAM at ≈30 Gbaud, implemented with a Silicon-Photonic (or LiNbO■, where thermal stability is paramount) IQ Mach-Zehnder Modulator (MZM) using the electro-optic effect**, because the baud-rate calculation shows DP-16QAM needs only ~30 Gbaud for 200 Gb/s (fitting a 50 GHz ITU-T G.698 grid slot across all 32 channels), its ~8 b/s/Hz ideal spectral efficiency comfortably exceeds the ≥4 b/s/Hz target, PAM4 would require an impractical ~100–120 Gbaud with much higher receiver bandwidth for the same bit rate, and the EO-driven MZI architecture is the only modulator combination offering the >35 GHz bandwidth, low chirp and amplitude/phase linearity that coherent DP-16QAM requires, at a footprint and thermal stability suited to a 32-channel transceiver array.

STAGE 3 – Optical Amplification Configuration (20 Marks)

3.1 Amplifier Technology Comparison

Amplifier	Gain Bandwidth	Saturated Output Power	Noise Figure (NF)	Gain Tilt / Crosstalk
EDFA	~35 nm (C-band), can extend with (+20 dBm class)	High (+20 dBm class)	Low (~4–6 dB)	Moderate gain tilt across band; needs GFF
SOA	Very broad (>80 nm possible)	Lower than EDFA	Higher (~6–9 dB)	Significant crosstalk from carrier-density no
Distributed/Lumped Raman	Tunable via pump wavelength	Distributed gain along fiber	Effective NF (distributed)	Low crosstalk, gain shape controllable by pum
Hybrid (EDFA + Raman)	Combines EDFA flat-band gain with Raman's distributed	High overall	Lowest overall (Raman pre-amplifies signal before EDFA)	Best overall tilt/crosstalk profile after GFF

3.2 Span Power-Budget Calculation

Given: input level −15 dBm/channel across 32 channels, target output +2 dBm/channel.

$$\textit{Required gain per channel} = P_{out} - P_{in} = 2\ \text{dBm} - (-15\ \text{dBm}) = \mathbf{17\ dB}$$

$$\textit{Total input power} = -15\ \text{dBm} + 10\cdot\log_{10}(32) = -15 + 15.05 \approx \mathbf{0.05\ dBm}\ (\approx 1.01\ \text{mW})$$

$$\textit{Total output power} = 2\ \text{dBm} + 10\cdot\log_{10}(32) = 2 + 15.05 \approx \mathbf{17.05\ dBm}\ (\approx 50.7\ \text{mW})$$

Since gain in dB is independent of the number of channels when all channels are scaled equally, the required **span gain is confirmed as 17 dB**, consistent with the per-channel figure and meeting the Stage-6 target of ≥17 dB/channel exactly.

3.3 Gain Non-Flatness and Gain Flattening Filters (GFF)

Across the ITU-T C-band grid, EDFA gain is inherently non-uniform (typically a few dB of ripple across 32–40 nm), which — left uncorrected — causes some channels to be over-amplified (risking nonlinear penalty) and others under-amplified (risking poor OSNR) after multiple cascaded spans. A Gain Flattening Filter (GFF), or dynamic power equalization using a wavelength-selective switch (WSS)/variable optical attenuator array, equalizes per-channel gain to within the target $\Delta G < 1.0$ dB specified in Stage 6, which is essential given that gain-tilt errors accumulate multiplicatively over N cascaded spans.

3.4 Cumulative OSNR Degradation Over N Cascaded Spans

$$\textit{1/OSNR_total(linear)} \approx \Sigma_i\ \textit{1/OSNR_i(linear)} \rightarrow \textit{for N identical spans: OSNR_total(dB)} \approx \textit{OSNR_1span(dB)} - 10\cdot\log_{10}(N)$$

For example, a single-span OSNR of 30 dB degrades to ~27 dB after 2 spans, ~24 dB after 4 spans, and ~20 dB after 10 spans — each doubling of span count costs roughly 3 dB of OSNR. Because ASE noise accumulates additively with every amplifier while signal power is restored each span, minimizing the number of discrete lumped-gain stages (by using distributed Raman gain along the fiber itself) directly reduces the effective ASE accumulation rate and extends reach for a fixed OSNR budget (>18 dB target).

Design Decision 2 – Optical Amplification Configuration

I recommend a **Hybrid EDFA + Distributed Raman amplification architecture with a Gain Flattening Filter (GFF) at each EDFA stage**, because the calculated 17 dB/channel gain requirement is achievable with headroom by either technology alone, but the hybrid combination minimizes the effective noise figure (Raman pre-amplifies the signal in the transmission fiber itself before it reaches the EDFA, reducing the ASE penalty the EDFA would otherwise add), keeps gain tilt within the required $\Delta G < 1.0$ dB via the GFF, and — per the cascaded-OSNR relation — gives the lowest cumulative OSNR degradation per span, which is essential for maintaining the >18 dB OSNR target over multiple amplified spans in a 32-channel, 6.4 Tb/s long-reach DWDM link.

STAGE 4 – WDM Impairment / DSP Compensation Configuration (20 Marks)

4.1 Physical Mechanisms of Key Impairments

- **Chromatic Dispersion (CD):** Different wavelength components travel at different group velocities in SMF-28, broadening pulses and causing inter-symbol interference (ISI) that grows with distance and baud rate.
- **Polarization Mode Dispersion (PMD):** Random fiber birefringence causes the two orthogonal polarization modes to travel at slightly different speeds, producing time-varying, statistically random pulse spreading and polarization rotation.
- **Self-Phase Modulation (SPM):** The Kerr nonlinearity makes a channel's own intensity modulate its own phase, causing nonlinear spectral broadening that interacts with CD.
- **Cross-Phase Modulation (XPM):** One WDM channel's intensity modulates the phase of neighbouring channels via the shared Kerr nonlinearity, causing inter-channel interference.
- **Four-Wave Mixing (FWM):** Three co-propagating wavelengths mix nonlinearly to generate a fourth frequency component, causing power depletion and crosstalk, worsened at low channel spacing and low local dispersion.
- **Polarization-Dependent Loss (PDL):** Component/fiber loss that varies with input polarization state, distorting the MIMO channel matrix seen by the coherent receiver.

4.2 Optical-Domain vs Digital-Domain Compensation

Aspect	Optical-Domain (DCF / Optical Phase Conjugation)	Digital-Domain (Coherent DSP)
CD compensation	Fixed, per-span fiber-based — adds loss and nonlinearity of its own	Fully adaptive, lossless (signal-processing only), handles any CD
Flexibility	Static — set for one design distance/dispersion map	Dynamically adaptive to changing network paths and reconfigurat
PMD	Cannot be compensated optically in a practical, adaptive way	MIMO adaptive equalization tracks and compensates PMD in rea
Footprint/cost	Adds physical fiber spools / conjugation modules, extra loss	No extra optical hardware; cost is in ASIC/DSP complexity and po
Nonlinearity handling	Limited; optical phase conjugation can partially invert nonlinear distortion	Digital backpropagation (DBP) can invert both CD and nonlinear

4.3 Coherent DSP Receiver Chain

- **1. Static CD equalization:** A frequency-domain (FFT-based) or long FIR filter removes the (known, slowly-varying) bulk chromatic dispersion accumulated over the link — implemented first because CD is large, linear, and largely static per route.
- **2. Adaptive polarization demultiplexing / PMD compensation (MIMO, e.g. CMA):** A 2×2 (or 4×4 real-valued) adaptive MIMO equalizer, typically trained with the Constant Modulus Algorithm (CMA) for QAM, separates the two transmitted polarizations, compensates PMD and residual CD, and tracks state-of-polarization rotation that varies on micro-to-millisecond timescales.
- **3. Carrier phase/frequency recovery:** Compensates laser phase noise and frequency offset between transmit and local-oscillator lasers, essential before symbol decision for any QAM format.
- **4. Digital Backpropagation (DBP), optional/selective:** Applies the inverse nonlinear Schrödinger equation (split-step Fourier method) to jointly undo fiber nonlinearity (SPM/XPM) and CD; gives the largest reach/Q-factor improvement under high launch power but at high computational cost (scales with number of steps/spans), so it is applied selectively (e.g., single-channel DBP, or low-complexity reduced-step DBP) where its Q-factor gain justifies the added ASIC power.
- **5. FEC decoding:** Soft-decision FEC (consistent with the <3.8×10■³ pre-FEC BER target) corrects residual errors to reach the required post-FEC error-free performance.

Design Decision 3 – DSP Compensation Configuration

I recommend a **fully digital coherent-DSP compensation chain — static frequency-domain CD equalization, adaptive CMA-based MIMO polarization demultiplexing/PMD compensation, carrier phase recovery, and low-complexity (reduced-step) Digital Backpropagation applied selectively — over optical-domain DCF/phase conjugation**, because DSP compensation is lossless, fully adaptive to varying link conditions (unlike a fixed DCF map), inherently required anyway for a coherent DP-16QAM receiver, and because DBP directly attacks the SPM/XPM nonlinear penalty that otherwise limits the achievable launch power and hence OSNR, giving the best achievable Q-factor/BER/reach for the lowest added optical hardware complexity.

STAGE 5 – Integrated WDM Transmission Strategy Evaluation (15 Marks)

5.1 Global Performance Summary

Parameter	Evaluated Value
Total aggregate throughput	6.4 Tb/s (32 × 200 Gb/s)
Spectral efficiency	~8 b/s/Hz ideal (DP-16QAM), well above ≥4 b/s/Hz target
Required span gain	17 dB/channel (matches Stage 3 calculation)
OSNR target	>18 dB, supported by hybrid EDFA+Raman low-NF design
DSP complexity	Full linear equalization + MIMO + selective DBP
Modulator bandwidth	>35 GHz (EO MZI, meets target)

5.2 Inter-Dependency Analysis

- **Modulation order → amplifier OSNR requirement:** Higher-order formats (DP-16QAM vs DP-QPSK) pack more bits/symbol into the same bandwidth but require roughly 4–6 dB higher OSNR for equivalent BER, directly tightening the amplifier noise-figure and span-count budget derived in Stage 3.
- **Modulation order → DSP complexity:** Higher-order QAM requires finer carrier-phase-noise tolerance and more MIMO equalizer taps/precision, increasing ASIC power — a direct trade-off against the spectral-efficiency gain.
- **MMI/AWG coupler losses → launch power:** Insertion loss in the multiplexer/demultiplexer directly subtracts from the available per-channel launch power budget calculated in Stage 3, meaning coupler loss must be included explicitly when setting the amplifier output-power target.
- **Amplifier spacing → nonlinear vs noise trade-off:** Shorter spans with lower per-span gain reduce nonlinear penalty but add more discrete noise-contributing stages; distributed Raman gain (Stage 3) mitigates this trade-off.

5.3 Reach / Channel-Density / Dynamic-Range Trade-off

Using DP-16QAM at 50 GHz channel spacing maximizes spectral efficiency and channel count (32 channels fit easily within the C-band) but has a lower nonlinear-noise tolerance and shorter reach than a lower-order format (e.g., DP-QPSK) at the same launch power; the hybrid Raman+EDFA amplification and selective DBP recover much of this reach penalty, so the chosen format remains viable for metro/regional distances while leaving headroom (via FEC and dynamic power equalization) to trade reach for channel count if a future upgrade to 64-QAM is required for even higher spectral efficiency over shorter links.

5.4 Critical Failure Points, Scalability, Standards Compliance

- **Failure points:** Excess gain tilt (if GFF under-performs) causing some channels to fall below BER threshold; residual PMD exceeding MIMO equalizer tracking speed under fast polarization transients; DBP complexity exceeding available ASIC power budget at full 32-channel scale.
- **Scalability:** The 50 GHz-grid, DP-16QAM architecture scales to more wavelength channels within the same C-band footprint, and to L-band extension using the same EDFA+Raman hybrid approach.
- **Standards compliance:** 50 GHz ITU-T G.698-compliant grid spacing, ~30 Gbaud per channel (within IEEE 802.3ck / OIF-CEI target baud-rate/interface ranges for 200G-per-lane coherent optics), and soft-decision FEC targeting the specified pre-FEC BER threshold together satisfy the stated standards.

Design Decision 4 – Integrated WDM Strategy

I recommend **adopting the complete integrated strategy — Coherent DP-16QAM/EO-MZI transmitters on a 50 GHz ITU-T G.698 grid, Hybrid EDFA+Raman amplification with GFF, and a full coherent DSP chain including selective DBP — for network deployment**, because the cross-stage evidence (6.4 Tb/s aggregate

capacity, ~8 b/s/Hz spectral efficiency, 17 dB/channel gain exactly meeting the power-budget calculation, low-NF hybrid amplification supporting the >18 dB OSNR target, and a DSP chain matched to every identified linear and nonlinear impairment) is internally consistent and meets every Stage-6 verification target simultaneously.

STAGE 6 – Final Verification Table (10 Marks)

Requirement / Performance Parameter	Specified / Target Value	Achieved / Evidence Value
Aggregate WDM capacity	6.4 Tb/s (32 ch × 200 Gb/s)	6.4 Tb/s — matches exactly (Stage 1)
Spectral efficiency & modulation choice	≥4 b/s/Hz (DP-16QAM)	~8 b/s/Hz ideal with DP-16QAM at ~30 Gbaud (Stage 2)
Modulator bandwidth & footprint	>35 GHz, compact PIC footprint	EO Si-photonic/LiNbO■ MZI, >35 GHz travelling-wave design (
Amplifier gain & gain flatness	≥17 dB/channel, ΔG < 1.0 dB	17 dB/channel calculated exactly; ΔG<1.0 dB via GFF (Stage 3)
OSNR / BER performance	OSNR > 18 dB, BER < 3.8×10■³ (pre-FEC)	Hybrid EDFA+Raman low-NF design supports >18 dB OSNR; s
DSP impairment mitigation	Full CD/PMD & nonlinear tolerance	Static CD EQ + MIMO/CMA + phase recovery + selective DBP
Standards & feasibility	ITU-T G.698 & IEEE 802.3ck compliance	50 GHz grid, ~30 Gbaud per channel, within target baud-rate ra

All achieved values meet or exceed their specified targets; the design is internally verified as consistent across Stages 1–5 with no unresolved deviation.

Final System Design (K6 – Create/Design)

Block-Level Architecture of the Complete DWDM Link

```
[32x CW Laser    [EO IQ-MZI       [Polarization    [MMI/AWG         [Hybrid Raman+EDFA     [AWG
 Array, ITU-T --> Modulators   --> Multiplexer  --> Multiplexer  --> Amplified Fiber    --> Demux   -->
 G.698 grid]      DP-16QAM]        (Pol-Mux)]       (32->1)]         Spans + GFF]          (1->32)]

     [Optical Isolators / Circulators / Faraday Rotators -- inserted at Tx output, each amplifier
      stage input/output, and Rx input for back-reflection and polarization control throughout]

  --> [Coherent Intradyne Coherent Receiver (ICR): 90-deg optical hybrid + Local Oscillator laser
       + Balanced Photodetectors] --> [High-Speed ADC, >=2x baud-rate sampling]

  --> [DSP Chain: Static CD Equalizer (FD-FIR) --> Adaptive MIMO Pol-Demux / PMD Comp. (CMA)
       --> Carrier Frequency & Phase Recovery --> Selective Digital Backpropagation (DBP)
       --> Symbol Decision] --> [Soft-Decision FEC Decoder] --> [200 Gb/s Client Data Out, x32 ch]
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Complete System Design Parameter Table

Parameter	Value
Carrier wavelengths	ITU-T C-band, 32 channels, 50 GHz grid (e.g., 1529.16–1561.83 nm region)
Grid spacing	50 GHz (ITU-T G.698-compliant)
Laser linewidth (Tx & LO)	Narrow linewidth, ■100 kHz class (ECL/DFB-based), to limit phase-noise penalty on 16-QAM
Modulation format / baud rate	Coherent DP-16QAM, ≈30 Gbaud/channel (incl. ~20% FEC overhead)
Modulator Vπ	Low-Vπ EO MZI (Si-photonic ~1–2 V travelling-wave; LiNbO■ ~3–5 V, design-dependent)
Modulator bandwidth	>35 GHz (3-dB electro-optic bandwidth)
Amplifier type / NF	Hybrid EDFA (NF ≈ 4–5 dB) + Distributed Raman pre-amplification
Amplifier gain / flatness	17 dB/channel, ΔG < 1.0 dB (via GFF)
Fiber type / attenuation (α)	SMF-28, α ≈ 0.2 dB/km
Fiber dispersion (D)	D ≈ 16–17 ps/nm/km (standard SMF-28 at 1550 nm)
Fiber nonlinear coefficient (γ)	γ ≈ 1.2–1.3 /W/km (standard SMF-28)
DSP filter taps (CD equalizer)	Sized to accumulated CD per span/route (frequency-domain implementation, scales with D×L)
MIMO equalizer	4×4 real-valued adaptive FIR, CMA pre-convergence + decision-directed tracking
FEC	Soft-decision FEC targeting pre-FEC BER < 3.8×10■³

Simulation Schematic Description (OptiSystem / OptSim / MATLAB)

A representative OptiSystem/OptSim testbed for this design would comprise: 32 CW laser sources on the 50 GHz grid → IQ MZM modulators driven by a DP-16QAM pattern/mapper → polarization multiplexer → AWG/MMI WDM multiplexer → fiber span blocks (SMF-28 with the α, D, γ values above) interleaved with hybrid Raman-pump + EDFA amplifier blocks and a GFF block → AWG demultiplexer → coherent receiver (90° hybrid + balanced PDs + LO laser) → ADC → the MATLAB/Octave-linked DSP chain (CD-EQ, CMA-MIMO, carrier recovery, selective DBP, FEC) → BER/Q-factor and constellation analyzers. Expected validating outputs include: open, well-separated 16-QAM constellation diagrams after DSP convergence; an eye diagram with high eye-opening prior to FEC; a WDM optical-spectrum plot confirming <1 dB per-channel gain flatness; and a Q-factor-vs-launch-power curve exhibiting the classic trade-off peak — Q-factor rising with launch power until ASE-limited, then falling as SPM/XPM nonlinear penalty dominates, with the optimum launch power identifying the recommended per-channel output power used in the Stage 3 budget.

Final Engineering Decision

WOULD YOU DEPLOY THE INVESTIGATED AND DESIGNED WDM SYSTEM?

YES – WITH TECHNICAL JUSTIFICATION

The investigated and designed DWDM system is recommended for deployment on the following consolidated technical grounds:

- **Modulation format efficiency and OSNR sensitivity:** Coherent DP-16QAM at ~30 Gbaud delivers ~8 b/s/Hz ideal spectral efficiency (double the ≥4 b/s/Hz target) while fitting comfortably in a 50 GHz grid slot; its OSNR requirement is met by the Stage 3 amplifier design with margin above the >18 dB target.
- **Photonic modulator integration, bandwidth and drive voltage:** The EO MZI architecture provides >35 GHz bandwidth with low, controllable chirp and a compact, integrable footprint suited to a 32-channel transceiver array, at practical drive voltages.
- **Amplification gain profile, noise accumulation and power budget:** The calculated 17 dB/channel gain exactly matches the specified −15 dBm → +2 dBm span requirement; the hybrid EDFA+Raman approach minimizes cumulative ASE/OSNR penalty across cascaded spans, and GFF keeps gain flatness within $\Delta G < 1.0$ dB.
- **DSP effectiveness in mitigating linear and nonlinear impairments:** The full coherent DSP chain (static CD-EQ, adaptive CMA-MIMO PMD/pol-demux, carrier recovery, selective DBP) directly addresses every impairment identified in Stage 4, with DBP specifically targeting the SPM/XPM nonlinear penalty that would otherwise cap reach and launch power.
- **Total system power consumption, scalability and cost-effectiveness:** Selective (reduced-complexity) DBP balances nonlinear-compensation gain against ASIC power; the Si-photonic/LiNbO■ EO-MZI choice and 50 GHz-grid architecture scale cleanly to more wavelengths or to L-band extension without redesign of the fundamental modulation/DSP strategy.

On this basis, the integrated DWDM transport system is technically justified, internally consistent across all six investigation stages, and satisfies every item in the Stage 6 verification table; it is therefore recommended for deployment, subject to standard prototype-stage OptiSystem/OptSim simulation confirmation of the exact Q-factor-vs-launch-power optimum and field validation of PMD-tracking speed under live-network conditions.

Assumptions: Where exact numerical values (e.g., laser linewidth, exact $V\pi$, precise DSP filter-tap counts, exact fiber length per span) were not explicitly specified in the scenario, representative, standards-consistent values from established DWDM engineering practice were assumed and stated at the point of use, consistent with the assignment's requirement to state assumptions wherever a parameter is not given.

References (representative): ITU-T Recommendation G.698.2; IEEE 802.3ck Task Force specifications; OIF-CEI-56G/112G implementation agreements; standard coherent-optical-communication and DWDM-system-engineering texts covering EDFA/Raman amplifier design and digital signal processing for coherent receivers.

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